Completeness of the Bethe Ansatz through Intersection Numbers in $\mathbb{P}^N$

Pei Wu
wupei5060309031@gmail.com

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Abstract

We develop an approach based on intersection numbers to study the completeness of the Bethe ansatz for periodic spin-1/2 XXX chains. Completeness of the Bethe ansatz predicts that the number of roots of the Bethe equations, after removing collisions, identifying permutations, and appropriately regularizing singular roots, equals $\binom{L}{N} - \binom{L}{N-1}$, where L is the length of the chain and N is the number of flipped spins. We propose a counting strategy that starts from the original Bethe equations and introduces a series of systems of polynomial equations as corrections. For each system, every root of the Bethe equations is assigned a local index given by its multiplicity. Our aim is that, after taking an appropriate weighted sum, this index equals $1/N!$ at physical roots and vanishes otherwise. Taking the global count, the weighted sum of intersection numbers would then yield the desired count. This claim is resolved conditionally: the case of roots with collisions among nonsingular components follows from the distinct-root assumption for the Q-polynomial satisfying the TQ relation. The case of unequal numbers of components at $\mathrm{i}/2$ and $-\mathrm{i}/2$ is resolved subsequently. For configurations containing singular pairs, we interpret the index geometrically through a deformation as a normalized count of nearby irreducible components with distinct coordinates and no site singular pair. It remains unclear why such components exist only when the number of components at each of $\mathrm{i}/2$ and $-\mathrm{i}/2$ is 1.

1 Background: the XXX Bethe ansatz

The Bethe ansatz originated in Bethe's 1931 study of the one-dimensional Heisenberg chain [3], where he constructed eigenfunctions using an ansatz for their coordinate dependence. The method and its extensions became central tools in the study of quantum integrable systems, including the XXX, XXZ, and XYZ spin chains and the related six- and eight-vertex models of statistical mechanics. Standard accounts of this broader setting include Baxter's book [4] and the treatment of the quantum inverse scattering method by Korepin, Bogoliubov, and Izergin [5].

Consider the periodic spin-1/2 XXX chain of length L, with state space $\mathcal{H} = (\mathbb{C}^2)^{\otimes L}$. One normalization of its Hamiltonian is

$$H = \frac{1}{4}\sum_{\ell=1}^{L}(\boldsymbol{\sigma}_\ell \cdot \boldsymbol{\sigma}_{\ell+1} - 1), \qquad \boldsymbol{\sigma}_{L+1} = \boldsymbol{\sigma}_1.$$

The sector with N down spins has dimension $\binom{L}{N}$. The state space carries a natural $\mathfrak{sl}_2$ action that commutes with H. The highest-weight subspace in the sector with N down spins has dimension

$$d(L,N) = \binom{L}{N} - \binom{L}{N-1}. \tag{1}$$

The algebraic Bethe ansatz constructs candidate eigenvectors

$$B(\lambda_1)\cdots B(\lambda_N)|0\rangle$$

from parameters satisfying

$$\left(\frac{\lambda_k+\mathrm{i}/2}{\lambda_k-\mathrm{i}/2}\right)^L=\prod_{j\neq k}\frac{\lambda_k-\lambda_j+\mathrm{i}}{\lambda_k-\lambda_j-\mathrm{i}},\qquad k=1,\dots,N. \tag{2}$$

Throughout this paper we assume $L\geq 2N$, where N is the number of flipped spins.

It is convenient to rescale the parameters by

$$\lambda_j=\frac{\mathrm{i}}{2}x_j,\qquad x_j=\frac{2\lambda_j}{\mathrm{i}}.$$

Define:

$$P(x_1,\dots,x_N)=(x_1+1)^L\prod_{j\neq 1}(x_1-x_j-2)-(x_1-1)^L\prod_{j\neq 1}(x_1-x_j+2). \tag{3}$$

We call P the *Bethe polynomial*; it is symmetric in its last $N-1$ arguments. Set

$$P_i(x)=P(x_i,x_1,\dots,\widehat{x_i},\dots,x_N).$$

The polynomial form of the periodic Bethe equations is $P_i=0$ for $i=1,\dots,N$.

Physical roots are required to have distinct coordinates; repeated coordinates are called *collisions*. Since the Bethe creation operator commutes, we identify configurations differing by a permutation. Roots containing both 1 and -1, which we may label as $x_1=1$ and $x_2=-1$, are called *singular*. When the remaining roots are distinct and different from ± 1, physical singular roots require the additional regularization condition

$$\left[-\prod_{j=3}^{N}\frac{x_j+1}{x_j-1}\right]^L=1. \tag{4}$$

See [6, 7] for regularization procedures.

By completeness of the Bethe ansatz, we mean that Bethe vectors obtained from physical roots, with singular roots appropriately regularized, span the highest-weight space. Early spectral results include Babbitt–Thomas's infinite-chain expansion [1] and Koma–Ezawa's two-down-spin analysis [2]. Combinatorial counts were developed by Kirillov [15] and Kirillov–Liskova [16], the latter assuming the string hypothesis. Baxter [17] examined exceptional roots and completeness for the six- and eight-vertex models. Algebraic-geometric approaches include Langlands–Saint-Aubin [18] and the quotient-ring counting methods of Jiang–Zhang [19].

For the homogeneous spin-1/2 XXX chain, Mukhin, Tarasov, and Varchenko [9, Section 12] established completeness in the polynomial-kernel formulation. The relation to physical roots was further clarified by Nepomechie–Wang's regularization criteria [6, 7], numerical root counts by Hao, Nepomechie, and Sommese [8], and Granet–Jacobsen's characterization of physical distinct-root configurations through polynomial companions and two zero-remainder conditions [10].

This note seeks an intersection-theoretic explanation of the completeness count. We express the expected count as a weighted combination of intersection numbers and study the associated local weights. The conjectural local and boundary identifications below concern this counting construction; completeness in the polynomial formulation is already established.

2 The counting strategy

Assume that the set of roots of $P_1 = \cdots = P_n = 0$ is finite, and write

$$X = \{x^1, \ldots, x^s\} \subseteq \mathbb{C}^n$$

for this set. Assign the target weights

$$w_i = \begin{cases} 1/n!, & x^i \text{ is a physical roots,} \\ 0, & \text{otherwise.} \end{cases}$$

As physical roots has distinct components, the desired completeness statement is

$$\sum_{i=1}^{s} w_i = d(L, n). \tag{5}$$

We work with formal sums of points with rational weights:

$$Z = \sum_i w_i(Z)[x^i], \qquad w_i(Z) \in \mathbb{Q}.$$

Here $[x^i]$ is a formal symbol for the point x^i. We write $\deg Z = \sum_i w_i(Z)$ for the total weight. For a map ϕ of point sets, we use the notation

$$\phi_*\left(\sum_p w_p[p]\right) = \sum_p w_p[\phi(p)],$$

adding weights when several points have the same image. In particular, passing from ordered tuples to unordered configurations simply adds the weights over each permutation orbit. We write $\mathrm{Sym}^r(\mathbb{A}^1)$ for the set of unordered configurations of r complex numbers, allowing repetitions.

For a zero-dimensional ideal $I \subseteq \mathbb{C}[x_1, \ldots, x_n]$, define the total number of its roots counted with multiplicity and the multiplicity at an isolated root p, respectively, by

$$\ell(I) = \dim_{\mathbb{C}}\big(\mathbb{C}[x_1, \ldots, x_n]/I\big),$$
$$\ell_p(I) = \dim_{\mathbb{C}}\big(\mathbb{C}[x_1, \ldots, x_n]/I\big)_{\mathfrak{m}_p},$$

where $\mathfrak{m}_p$ is the maximal ideal of the point p, and $\ell_p(I) = 0$ if $p \notin V(I)$. The associated formal sum is

$$Z(I) = \sum_{p \in V(I)} \ell_p(I)[p]$$

and

$$\deg Z(I) = \ell(I).$$

Our strategy is to seek ideals I_j and weights $w_j^{\mathrm{aux}} \in \mathbb{Q}$ such that each I_j is generated by n polynomials, is zero-dimensional, and satisfies $V(I_j) \subseteq X$. Set $Z_j = Z(I_j)$ and write

$$Z_j = \sum_{i=1}^{s} l_{ij}[x^i], \qquad l_{ij} = \ell_{x^i}(I_j),$$

with $l_{ij} = 0$ when $x^i \notin V(I_j)$. We require the formal sum

$$Z_{\mathrm{Bethe}} = \sum_j w_j^{\mathrm{aux}} Z_j \tag{6}$$

to equal $\sum_i w_i[x^i]$; equivalently,

$$\sum_j w_j^{\mathrm{aux}} l_{ij} = w_i \qquad \text{for every } i. \tag{7}$$

Taking degrees then gives

$$\sum_i w_i = \sum_j w_j^{\mathrm{aux}} \ell(I_j), \qquad \ell(I_j) = \deg Z_j = \sum_i l_{ij}. \tag{8}$$

The generators of I_j allow us to study their global intersection numbers. We defer the compactification in $\mathbb{P}^n$ and the analysis of contributions at infinity to a later section.

The proposed approach has two steps:

1. To remove collision and permutation redundancy we make use of Möbius inversion on the partition lattice. We call this process *depermutation*.

2. Construct an excess formal sum on the singular-pair locus and subtract it. The resulting formal sum should satisfy the pointwise identities (7) and hence the degree identity (5).

The first step supplies explicit formal sums Z_j and rational weights w_j^{aux} for depermutation. The second modifies these weights and adds formal sums on the singular locus so that the resulting local sums are the target weights w_i. The advantage is the passage from the local identities (7) to the global calculation (8).

We construct auxiliary ideals from the Bethe polynomials P_i. Let $A, B \subseteq [n]$ be disjoint, with either $A = B = \varnothing$ or both A and B nonempty. Let $\pi = \{S_1, \dots, S_m\}$ be a partition of $R = [n] \setminus (A \cup B)$, and choose a representative $r_\alpha \in S_\alpha$ for each block. For $t \in R$, define the reduced polynomial by

$$P_t^{\mathrm{red}}(x_R) := \frac{P_t(x)|_{x_a=1\ (a\in A),\, x_b=-1\ (b\in B)}}{(x_t+1)^{|A|}(x_t-1)^{|B|}}. \tag{9}$$

This is well-defined: the two terms in the numerator are both divisible by $(x_t+1)^{|A|}(x_t-1)^{|B|}$, since $L \geq 2n$. For $A = B = \varnothing$, set $P_t^{\mathrm{red}} = P_t$. The dependence on A, B is suppressed in the notation. Define

$$\begin{aligned} I_{A,B;\pi} = \big(&x_a - 1 \ (a \in A), \quad x_b + 1 \ (b \in B), \\ &P_{r_\alpha}^{\mathrm{red}} \ (1 \leq \alpha \leq m), \\ &x_t - x_{r_\alpha} \quad (1 \leq \alpha \leq m,\ t \in S_\alpha \setminus \{r_\alpha\})\big). \end{aligned} \tag{10}$$

The support is contained in the locus in X with the specified fixed roots and block equalities. The reduced equations impose additional conditions on that locus. The ideal is independent of the chosen representatives. Let

$$Z_{A,B;\pi} = Z(I_{A,B;\pi})$$

assuming $I_{A,B;\pi}$ is zero-dimensional. We abbreviate $Z_{\varnothing,\varnothing;\pi}$ to Z_π.

Our aim is to find rational weights $w_{A,B;\pi}^{\mathrm{aux}}$ such that

$$Z_{\mathrm{Bethe}} = \sum_{A,B,\pi} w_{A,B;\pi}^{\mathrm{aux}} Z_{A,B;\pi}.$$

2.1 Depermutation

The general principle for removing collisions is Möbius inversion on the partition lattice; see [11, Section 3.7]. To illustrate it, let Ω be a finite set and $S \subseteq \Omega^n$ be invariant under coordinate permutations. For a partition π of $[n]$, let S_π consist of the tuples in S satisfying $x_i = x_j$ whenever i, j belong to the same block of π; coordinates in different blocks are still allowed to coincide. Order the partition lattice Π_n by refinement, with $\hat{0}$ the partition into singletons. Then the number of permutation orbits in S with pairwise distinct coordinates is

$$\frac{1}{n!} \sum_{\pi \in \Pi_n} \mu(\hat{0}, \pi)\, \#S_\pi, \qquad \mu(\hat{0}, \pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!.$$

We now replace these finite sets by formal sums of points recording local multiplicities. Consider a general polynomial $F(x_1, \dots, x_n)$ symmetric in its last $n-1$ arguments. Set

$$F_i(x) = F(x_i, x_1, \dots, \widehat{x_i}, \dots, x_n).$$

For a partition $\pi = \{S_1, \dots, S_m\}$ choose a representative $r_a \in S_a$ for each block, and set

$$I_{F;\pi} = \big(F_{r_a},\ x_b - x_{r_a} : 1 \le a \le m,\ b \in S_a \setminus \{r_a\}\big),$$

and set $Z_{F;\pi} = Z(I_{F;\pi})$.

Definition 2.1. The *depermutation sum* of F is

$$Z_F^{\mathrm{dep}} = \frac{1}{n!} \sum_{\pi \in \Pi_n} \mu(\hat{0}, \pi) Z_{F;\pi}. \tag{11}$$

We now return to the XXX polynomial $F = P$. The Z_P^{dep} need further correction from singular roots. To formulate this singular correction, extend depermutation to systems with fixed singular roots. For $0 \le k \le \lfloor n/2 \rfloor$, let $A, B \subseteq [n]$ be disjoint subsets with $|A| = |B| = k$, and put $R = [n] \setminus (A \cup B)$ and $r = |R| = n - 2k$. Set

$$\begin{aligned} Z_{A,B}^{\mathrm{dep}} &= \frac{1}{r!} \sum_{\pi \in \Pi(R)} \mu(\hat{0}, \pi) Z_{A,B;\pi}, \\ Z_k^{\mathrm{dep}} &= \frac{(k!)^2 (n-2k)!}{n!} \sum_{\substack{A,B \subseteq [n],\ A \cap B = \varnothing \\ |A| = |B| = k}} Z_{A,B}^{\mathrm{dep}}. \end{aligned} \tag{12}$$

Here $\Pi(R)$ is the partition lattice of the remaining labels; depermutation acts only on the r free coordinates. In particular, $Z_0^{\mathrm{dep}} = Z_P^{\mathrm{dep}}$.

Conjecture 2.2 (Alternating completeness formula)**.** *Assume $L \ge 2n$ and that the reduced partition ideals are zero-dimensional. The formal sum*

$$\boxed{\tilde{Z} = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{L}{k} Z_k^{\mathrm{dep}}} \tag{13}$$

equals Z_{Bethe}.

The projective intersection number is computed in Section 2.3. Its identification with the affine degree and the target physical weights requires the boundary and local arguments described below.

Remark 2.3 (Sums of functions of Bethe roots)**.** Many computations require sums of functions evaluated at Bethe roots, rather than just their number. The proposed pointwise weight identity extends directly to such sums. Let h be a symmetric polynomial on n variables. If Conjecture 2.2 holds, then

$$\sum_{[p]\text{ physical}} h(p) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{L}{k} \sum_{p \in X} w_k(p) h(p). \tag{14}$$

Here $w_k(p)$ is the weight of the ordered point p in Z_k^{dep}. The left-hand sum contains one representative of each physical permutation orbit, whereas the right-hand inner sum runs over all ordered points of X. Thus the weights of all ordered representatives are added before passing to an orbit. The counting formula is the case $h = 1$. Expanding the depermutation sums expresses the right-hand side in terms of the sums over $Z_{A,B;\pi}$.

The sums over $Z_{A,B;\pi}$ can be computed numerically in the finite quotient algebra $A_j = \mathbb{C}[x_1, \ldots, x_n]/I_j$ as

$$\sum_{p \in V(I_j)} \ell_p(I_j) h(p) = \mathrm{Tr}_{A_j}(M_h), \qquad M_h(a) = ha.$$

The same formula applies to rational h whose denominator is nonzero at every auxiliary root, see [19]. One also expect that in some case the sum can be decided explicitly, say for small N.

2.2 A generalized TQ relation and depermutation

Fix a polynomial F symmetric in its last $n-1$ arguments, and write $F_i(x) = F(x_i, x_1, \ldots, \widehat{x_i}, \ldots, x_n)$. Let e_j and $\hat{e}_j$ denote the elementary symmetric polynomials in $x_1, \ldots, x_n$ and $x_2, \ldots, x_n$, respectively, and set $e_0 = 1$. The map from ordered roots to their elementary symmetric functions is

$$q : \mathbb{A}^n \longrightarrow \mathrm{Sym}^n(\mathbb{A}^1) \simeq \mathbb{A}^n, \qquad x \longmapsto e(x).$$

Using $e_1, \ldots, e_n$ as coordinates on the target, define

$$Q(t; e) = \sum_{j=0}^{n} (-1)^j e_j t^{n-j}.$$

Write Z^{dep} for the depermutation sum of this system.

The TQ relation transforms equations in the roots x into equations in their elementary symmetric functions e. We construct a polynomial $\Phi(t; e)$ whose divisibility by $Q(t; e)$ encodes the equations at distinct roots. By symmetry, there is a polynomial g such that

$$F(x_1, \ldots, x_n) = g(x_1; \hat{e}_1, \ldots, \hat{e}_{n-1}).$$

Set

$$E_j(t; e) = \sum_{k=0}^{j} (-t)^k e_{j-k}, \qquad 1 \le j < n,$$
$$\Phi(t; e) = g\big(t; E_1(t; e), \ldots, E_{n-1}(t; e)\big).$$

The elementary symmetric identities give $E_j(x_1; e(x)) = \hat{e}_j$, hence

$$\Phi(x_1; e(x)) = g(x_1; \hat{e}_1, \ldots, \hat{e}_{n-1}) = F_1(x).$$

If the coordinates are distinct, the equations $F_i(x) = 0$ therefore give $\Phi(x_i; e(x)) = 0$ for every i, by symmetry. Thus they are equivalent to $Q \mid \Phi$ on this locus.

Since Q is monic, polynomial division gives unique polynomials U, R with coefficients in $\mathbb{C}[e_1, \ldots, e_n]$ such that

$$\Phi(t; e) = U(t; e)Q(t; e) + R(t; e), \qquad R(t; e) = \sum_{j=0}^{n-1} \rho_j(e)t^j. \tag{15}$$

Define the *generalized TQ ideal*

$$J = (\rho_0, \ldots, \rho_{n-1}) \subseteq \mathbb{C}[e_1, \ldots, e_n]. \tag{16}$$

Proposition 2.4 (Simple zeros with repeated coordinates)**.** *Let $I_F = (F_1, \ldots, F_n)$, and let $p \in V(I_F)$ be a simple zero, meaning that the Jacobian $(\partial F_i/\partial x_j)(p)$ is invertible. If $p_a = p_b$ for some $a \neq b$, then $q(p) \notin V(J)$. In particular, when J is zero-dimensional, the weight of $q(p)$ in $Z(J)$ is zero.*

Proof. Put $e^* = q(p)$ and $u = p_a = p_b$, and let v be the nonzero tangent vector with $v_a = 1$, $v_b = -1$, and all other entries zero. Since each elementary symmetric polynomial is invariant under interchanging x_a and x_b, its two corresponding partial derivatives agree at p. Hence $dq_p(v) = 0$.

The construction of Φ gives the polynomial identities $F_i(x) = \Phi(x_i; q(x))$ for all i, including on the collision locus. Differentiating along v therefore gives

$$(dF_i)_p(v) = \partial_t \Phi(p_i; e^*)v_i.$$

Suppose $e^* \in V(J)$. Then $Q(t; e^*) \mid \Phi(t; e^*)$. Because $(t-u)^2 \mid Q(t; e^*)$, we have $\partial_t \Phi(u; e^*) = 0$. The displayed derivatives thus vanish for $i = a, b$, and also for every other i since $v_i = 0$. Consequently v lies in the kernel of the Jacobian of I_F at p, contradicting its invertibility. Thus $q(p) \notin V(J)$, so $\ell_{q(p)}(J) = 0$ whenever the local multiplicity is defined as above. □

The formal sum associated to the TQ ideal is the pushforward of the depermutation sum to $\mathrm{Sym}^n(\mathbb{A}^1)$:

$$q_* Z^{\mathrm{dep}} = Z(J).$$

Here we assume the relevant ideals are zero-dimensional. Informally, the idea is to perturb F so that all zeros are simple.

For the XXX system, we use the following additional input.

Distinct-root assumption away from the singular values. Let $L \geq 2n$, and suppose a monic polynomial Q of degree n admits a polynomial T satisfying Baxter's relation

$$T(t)Q(t) = (t+1)^L Q(t-2) + (t-1)^L Q(t+2). \tag{17}$$

Write

$$Q(t) = (t-1)^a (t+1)^b Q_{\mathrm{reg}}(t), \qquad Q_{\mathrm{reg}}(1)Q_{\mathrm{reg}}(-1) \neq 0.$$

We assume that Q_{reg} is squarefree. Equivalently, every root of Q outside $\{1, -1\}$ is simple; repetitions at the singular values are allowed.

The exclusion of repeated roots through polynomiality of T is discussed in [12, Introduction, eq. (1.3) and footnote 2], where numerical evidence is reported and a rigorous proof for the spin-1/2 XXX chain is stated to be unavailable to the authors. We use the precise version above, including configurations with singular roots, as an assumption rather than a proved result. Together with the pushforward identity, it rules out collisions away from ± 1 in the support of Z^{dep}. The same conclusion holds for the reduced balanced fixed-pair systems: if H satisfies the TQ relation after fixing k pairs, then $Q = (t^2-1)^k H$ satisfies (17).

2.3 Projective intersection numbers and the combinatorial identity

We compactify the ordered root space by

$$\mathbb{A}^n \hookrightarrow \mathbb{P}^n, \qquad x \longmapsto [1 : x_1 : \cdots : x_n].$$

Write $[z : x_1 : \cdots : x_n]$ for homogeneous coordinates; $z = 0$ is the boundary. The homogenization of P is

$$\hat{P}(z, x_1, \ldots, x_n) = \frac{1}{z}\big((x_1 + z)^L \prod_{j\neq 1}(x_1 - x_j - 2z) - (x_1 - z)^L \prod_{j\neq 1}(x_1 - x_j + 2z)\big)$$

This is a polynomial of degree $L + n - 2$. The reduced Bethe polynomials are homogenized similarly. We compute the projective intersection number associated with $Z_{A,B;\pi}$ in this compactification [13].

Definition 2.5 (Projective intersection number)**.** Put $r = n - |A| - |B|$. For a partition π of $[n] \setminus (A \cup B)$ with m blocks, set

$$\boxed{\deg_{\mathbb{P}^n} Z_{A,B;\pi} = (L + r - 2)^m.} \tag{18}$$

Below, $\deg_{\mathbb{P}^n}$ denotes this intersection number, extended linearly to weighted sums. The notation deg continues to denote affine total weight. We will show that the projective formal sum $\tilde{Z}$ has zero weight at every boundary point.

Remark 2.6 (Agreement of BKK theorem)**.** The intersection number has the following BKK interpretation [14, Chapter 3]. Put $r = n-|A|-|B|$ and $D = L+r-2$. For $r \geq 1$, retain the ambient monomial supports of the reduced equations when identifying coordinates within each block, before combining coefficients that may cancel. Let $\mathcal{N}_\alpha \subseteq \mathbb{R}^m_{\geq 0}$ be the downward closure of the convex hull of the retained exponents for block α. The ambient equation has total degree at most D and a nonzero pure term $x^D_{r_\alpha}$. Consequently,

$$[0, De_\alpha] \subseteq \mathcal{N}_\alpha \subseteq D\Delta_m, \qquad \Delta_m = \{u \in \mathbb{R}^m_{\geq 0} : \textstyle\sum_\alpha u_\alpha \leq 1\}.$$

Normalize mixed volume by $\mathrm{MV}(\Delta_m, \ldots, \Delta_m) = 1$. Monotonicity gives

$$\begin{aligned} D^m &= \mathrm{MV}([0, De_1], \ldots, [0, De_m]) \\ &\leq \mathrm{MV}(\mathcal{N}_1, \ldots, \mathcal{N}_m) \leq \mathrm{MV}(D\Delta_m, \ldots, D\Delta_m) = D^m. \end{aligned}$$

Thus the mixed volume equals D^m, agreeing with the intersection number in the projective compactification.

Proposition 2.7. *For $|A| = |B| = k$ and $r = n - 2k$,*

$$\deg_{\mathbb{P}^n} Z^{\mathrm{dep}}_{A,B} = \deg_{\mathbb{P}^n} Z^{\mathrm{dep}}_k = \binom{L + r - 2}{r}. \tag{19}$$

Consequently,

$$\boxed{\deg_{\mathbb{P}^n} \tilde{Z} = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{L}{k}\binom{L + n - 2k - 2}{n - 2k} = \binom{L}{n} - \binom{L}{n-1}.} \tag{20}$$

Proof. The partition-lattice identity

$$\sum_{\pi\in\Pi_r} \mu(\hat{0},\pi)D^{|\pi|} = D(D-1)\cdots(D-r+1)$$

follows by counting injective maps from $[r]$ to a set of D elements: there are $D^{|\pi|}$ maps constant on each block of π, and Möbius inversion removes all coincidences. Dividing by $r!$ gives

$$\deg_{\mathbb{P}^n} Z^{\mathrm{dep}}_{A,B} = \frac{1}{r!}\sum_{\pi\in\Pi_r} \mu(\hat{0},\pi)D^{|\pi|} = \binom{D}{r} = \binom{L+r-2}{r}.$$

For $r=0$, both sides are one. The average defining Z^{dep}_k has $n!/((k!)^2 r!)$ equal-degree terms, so its projective intersection number is the same.

For the final identity, use formal power series:

$$\left(\sum_{k=0}^{L}(-1)^k\binom{L}{k}t^{2k}\right)\left(\sum_{r\geq 0}\binom{L+r-2}{r}t^r\right) = \frac{(1-t^2)^L}{(1-t)^{L-1}}$$
$$= (1-t)(1+t)^L.$$

Taking the coefficient of t^n proves (20), with $\binom{L}{-1} = 0$ if $n=0$. □

Thus the projective intersection calculation gives the desired number. To prove Conjecture 2.2, we must connect this projective intersection number with the affine roots and establish the prescribed weight at every point.

2.4 Plan for a proof of Conjecture 2.2

For $p \in X$, define its exact singular sets

$$A(p) = \{j : p_j = 1\}, \qquad B(p) = \{j : p_j = -1\},$$

and let $\pi(p)$ be the equality partition of the remaining labels: two labels belong to the same block precisely when their coordinates are equal. Write $a = |A(p)|$ and $b = |B(p)|$. If one of a, b is positive, the original Bethe equation at that fixed root forces the other to be positive as well. Hence either $a = b = 0$ or $a, b \geq 1$. These exact labels classify the affine configurations. Write $w_k(p)$ for the weight of p in Z^{dep}_k, and set $w(p) = w_i$ when $p = x^i$.

The following steps are needed to prove the conjecture:

1. **Boundary points.** Show that every point on the boundary has zero weight in $\tilde{Z}$. Thus the total weight of $\tilde{Z}$ is supported entirely in the affine chart.

2. Every Z^{dep}_k has zero weight at configurations with collisions outside ± 1, so its support has $\pi(p) = \hat{0}$.

3. Every Z^{dep}_k contains only points with $a = b$.

4. For distinct nonsingular roots, establish local multiplicity one for $(P_1, \ldots, P_n)$ and a nonzero Bethe vector. Only $k = 0$ contributes, giving weight $1/n!$.

5. **Exactly one singular pair: $a = b = 1$ and $\pi = \hat{0}$.** Only $k = 0, 1$ can contribute, so prove

$$w_0(p) - Lw_1(p) = w(p) \tag{21}$$

where $w(p)$ is the target weight above. Writing $e_p = \ell_p(I_{A(p),B(p);\hat{0}})$ for the local multiplicity of the reduced fixed-pair system, the precise local-multiplicity claim is

$$\ell_p((P_1,\dots,P_n)) = \begin{cases} Le_p + 1, & p \text{ satisfies the physical condition (4)}, \\ Le_p, & \text{otherwise.} \end{cases} \tag{22}$$

For a simple reduced root, $e_p = 1$.

6. **Multiple balanced singular pairs: $a = b \geq 2$ and $\pi = \hat{0}$.** These points have repeated coordinates and hence target weight zero. Prove

$$\sum_{k=0}^{a} (-1)^k \binom{L}{k} w_k(p) = 0,$$

Reduction using the distinct-root assumption. The distinct-root assumption substantially reduces these tasks. Together with the TQ interpretation of depermutation, it excludes every affine configuration with a collision outside ± 1, for the full system and for each reduced balanced fixed-pair system. Thus only configurations with $\pi(p) = \hat{0}$ need further consideration.

At infinity, Section 3 proves directly that every boundary root has two equal nonzero homogeneous root coordinates. This locates the boundary support on collision diagonals. It does not by itself prove cancellation of the local multiplicities; that requires a comparison of the full and partition intersections at the boundary.

For the remaining affine configurations, the TQ relation also excludes different numbers of roots at the two singular values, $a \neq b$, by comparing vanishing orders at 1 and -1. The same argument applies after fixing any k balanced pairs, so $w_k(p) = 0$ for every $0 \leq k \leq \min(a,b)$ at an unbalanced configuration. Consequently, among singular affine configurations, only the balanced cases $a = b$ remain: the physical-weight identity for $a = b = 1$ and the alternating cancellation for $a = b \geq 2$.

3 Roots at infinity

We first locate the boundary roots of the projective equations. Use coordinates $[z : X_1 : \cdots : X_n]$ on $\mathbb{P}^n$, with $x_i = X_i/z$ in the affine chart. The homogenization of P_i to degree $L+n-2$ is

$$\widehat{P}_i(z,X) = \frac{1}{z}\left((X_i+z)^L \prod_{j\neq i}(X_i - X_j - 2z) - (X_i - z)^L \prod_{j\neq i}(X_i - X_j + 2z) \right).$$

The numerator vanishes identically at $z = 0$, so this is a polynomial. Expanding its numerator to first order in z gives the boundary equations

$$\widehat{P}_i(0,X) = 2X_i^{L-1}\left(L\prod_{j\neq i}(X_i - X_j) - 2X_i \sum_{j\neq i}\prod_{h\neq i,j}(X_i - X_h) \right). \tag{23}$$

Proposition 3.1 (Collisions at infinity)**.** *Assume $L \geq 2n$ and $n \geq 1$. Every common zero $[0 : X_1 : \cdots : X_n]$ of the projective Bethe equations has*

$$X_i = X_j \neq 0 \quad \textit{for some distinct } i,j.$$

In particular, every boundary root lies on a collision diagonal.

Proof. Suppose the nonzero coordinates were pairwise distinct. Let $S = \{i : X_i \neq 0\}$ and $s = |S| \geq 1$. For $i \in S$ all factors $X_i - X_j$ are nonzero, so dividing (23) by $2X_i^{L-1}\prod_{j\neq i}(X_i - X_j)$ gives

$$L = 2\sum_{j\neq i}\frac{X_i}{X_i - X_j} = 2(n-s) + 2\sum_{\substack{j\in S\\ j\neq i}}\frac{X_i}{X_i - X_j}.$$

Sum over $i \in S$. Each unordered pair contributes one to the double sum, because

$$\frac{X_i}{X_i - X_j} + \frac{X_j}{X_j - X_i} = 1.$$

It follows that

$$Ls = 2s(n-s) + s(s-1), \qquad L = 2n - s - 1 \leq 2n - 2,$$

contradicting $L \geq 2n$. □

The argument also applies after fixing k balanced singular pairs. There are $r = n - 2k$ free roots, and their reduced equations are

$$\begin{aligned}(x_i+1)^{L-k}(x_i-3)^k\prod_{j\neq i}(x_i - x_j - 2)\\ -(x_i-1)^{L-k}(x_i+3)^k\prod_{j\neq i}(x_i - x_j + 2) = 0,\end{aligned}$$

where the products run over the other free roots. Homogenize them to degree $L+r-2$ in the free-root space $\mathbb{P}^r$. Their leading forms are (23) with r free coordinates and $L - 4k$ in place of the coefficient L inside the parentheses. The prefactor remains $2X_i^{L-1}$. Since $L - 4k \geq 2r$, the same proof forces two equal nonzero free-root coordinates at every boundary root. For $r = 1$ there are no boundary roots; for $r = 0$ there is no free-root boundary.

This collision statement uses no distinct-root assumption. However, support on collision diagonals is weaker than vanishing of the depermutation weight. The homogenized equations may have nontrivial local multiplicities, or even positive-dimensional intersections at infinity. Comparing those contributions across the partition systems, and proving their cancellation, remains necessary to identify the projective intersection count with the affine total weight.

4 Exactly one singular pair: $a = b = 1$

The case of one singular pair has been studied through regularization of the Bethe vector by Nepomechie and Wang [6, Section 3], including twist regularization [7, Section 2]. Granet and Jacobsen [10, Sections 2–4] relate physical singular roots to polynomial companions and additional zero-remainder conditions. Here we study the same configurations through their multiplicities in the full and reduced polynomial systems.

Let $n \geq 2$, $L \geq 2n$, and let $p = (1, -1, z_3, \ldots, z_n)$ have pairwise distinct coordinates. Set

$$I_{\mathrm{full}} = (P_1, \ldots, P_n), \qquad I_{\mathrm{red}} = I_{\{1\},\{2\};\hat{0}}.$$

We prove a weaker version of (22):

Proposition 4.1. *Suppose $p = (1, -1, z_3, \ldots, z_n)$ is a simple zero of I_{red}. Then $\ell_p(I_{\mathrm{full}}) \geq L + 1$ if p satisfies (4) and $\ell_p(I_{\mathrm{full}}) = L$ otherwise.*

Proof. Write $x_1 = 1 + u$, $x_2 = -1 + v$, and $x_j = y_j$ for $j \geq 3$. Define

$$C(u, y) = \prod_{j=3}^{n}(u - 1 - y_j), \qquad D(u, y) = \prod_{j=3}^{n}(u + 3 - y_j).$$

Since $C(0, z) \neq 0$, the equation $P_1 = 0$ determines

$$v = u - \frac{4u^L D(u, y)}{(2 + u)^L C(u, y) - u^L D(u, y)} = u + O(u^L).$$

After this substitution, write $P_2 = u^L H(u, y)$ and $P_j = G_j(u, y)$ for $3 \leq j \leq n$. These functions are regular near $(u, y) = (0, z)$. At $u = 0$, $G_j(0, y) = (y_j^2 - 1)P_j^{\mathrm{red}}(y)$. The Jacobian of $(G_3, \ldots, G_n)$ in the y variables is invertible at $(0, z)$: its rows are the reduced Jacobian rows multiplied by the nonzero factors $z_j^2 - 1$. The implicit function theorem therefore gives a unique local curve $y = y(u)$ solving these equations. The full local algebra becomes

$$\mathbb{C}[[u]]/(u^L h(u)), \qquad h(u) = H(u, y(u)).$$

To determine $h(0)$, direct substitution gives

$$H(0, y) = 4(-1)^{n-2}\left[-\prod_{j=3}^{n}(y_j + 3) + (-1)^L \prod_{j=3}^{n}\frac{(y_j - 1)(y_j - 3)}{y_j + 1}\right].$$

A distinct-root root has no exact strings other than the singular pair; in particular, $z_j \neq \pm 3$ and $z_i - z_j \neq \pm 2$ for $i, j \geq 3$; see [10, Lemma 1]. Multiplying the reduced equations therefore gives

$$\left(\prod_{j=3}^{n}\frac{z_j + 1}{z_j - 1}\right)^{L-1} = \prod_{j=3}^{n}\frac{z_j + 3}{z_j - 3}.$$

Consequently,

$$h(0) = 0 \quad \Longleftrightarrow \quad \left[-\prod_{j=3}^{n}\frac{z_j + 1}{z_j - 1}\right]^L = 1. \tag{24}$$

If p is nonphysical, then $h(0) \neq 0$ and the multiplicity is L. Otherwise, $h(0) = 0$ and the multiplicity is greater than L, proving the claim.

□

5 The local index as a degree of remaining components

We give a geometric interpretation of the alternating sum using the original partition ideals. The argument uses an inhomogeneous family and two inclusion–exclusion operations: depermutation removes collisions, and the alternating sum removes components carrying a site singular pair. The conclusion is conditional on the genericity assumptions stated below; no TQ relation is needed.

The family and its fixed-pair systems. Introduce site parameters $\theta_1, \ldots, \theta_L$ and put

$$a_\theta(t) = \prod_{j=1}^{L}(t + \theta_j + 1), \qquad d_\theta(t) = \prod_{j=1}^{L}(t + \theta_j - 1).$$

The deformed original equations are

$$P_i^\theta(x) = a_\theta(x_i)\prod_{s\neq i}(x_i - x_s - 2) - d_\theta(x_i)\prod_{s\neq i}(x_i - x_s + 2) = 0.$$

They specialize to $P_i = 0$ at $\theta = 0$. Put

$$T = \operatorname{Spec}\mathbb{C}[[\theta_1, \ldots, \theta_L]],$$

and work in $T \times \mathbb{A}^n$. We assume that the relevant local families are finite and flat over T. Barred symbols $\bar{Z}$ denote the corresponding formal sums of families, retaining their scheme-theoretic multiplicities. Their specialization at $\theta = 0$ is the formal sum of points associated to the special fiber. Write $\bar{X}$ for the set of irreducible components of the deformed solution locus $P_1^\theta = \cdots = P_n^\theta = 0$. We assume that the deformed Bethe scheme is generically reduced along every irreducible component dominating T. Since the family is finite and the base field has characteristic zero, this means that its geometric generic fiber consists of simple zeros: the Jacobian of the n original equations in the n root variables is invertible at every generic solution. The homogeneous special fiber may still have nontrivial multiplicities.

Work locally at a homogeneous solution

$$p = (1^k, -1^k, z_1, \ldots, z_{n-2k}), \qquad z_i \neq \pm 1, \quad z_i \neq z_j \ (i \neq j).$$

Definition 5.1 (Labels and equality partitions of a component)**.** Let $C \in \bar{X}$ be a component of the local family through $(0, p)$ that dominates T. Define

$$f_C, g_C : [k] \longrightarrow [L] \cup \{\mathcal{E}\}$$

by

$$f_C(a) = \begin{cases} j, & x_a = 1 - \theta_j \text{ identically on } C, \\ \mathcal{E}, & \text{no such } j \text{ exists}, \end{cases}$$

$$g_C(b) = \begin{cases} j, & x_{k+b} = -1 - \theta_j \text{ identically on } C, \\ \mathcal{E}, & \text{no such } j \text{ exists}. \end{cases}$$

These identities are taken in the function field of C. Since C dominates T, distinct site parameters remain distinct generically, so a site label is unique. The symbol $\mathcal{E}$ records a coordinate that specializes to 1 or -1 but is not pinned to a site endpoint along C.

Put $E_C^+ = f_C^{-1}(\mathcal{E})$ and $E_C^- = g_C^{-1}(\mathcal{E})$. Let $\pi_1(C)$ be the partition of E_C^+ in which a, a' belong to the same block exactly when $x_a = x_{a'}$ identically on C. Define $\pi_2(C)$ on E_C^- in the same way, using the coordinates x_{k+b}. Call C *exceptional* if

$$f_C = g_C \equiv \mathcal{E}, \qquad \pi_1(C) = \pi_2(C) = \hat{0},$$

where $\hat{0}$ denotes the partition into singletons.

Let $K = \operatorname{Frac}\mathbb{C}[[\theta_1, \ldots, \theta_L]]$ and let Γ_p be the finite set of geometric generic solutions in the local family specializing to p. For a component C, put

$$J_C = f_C([k]) \cap [L] = g_C([k]) \cap [L], \qquad m_C = |J_C|, \qquad d_C = [K(C) : K].$$

The two site images agree because the equation at a pinned endpoint forces its partner at the same site. Define

$$D_m(p) = \sum_{\substack{C \text{ has generically distinct coordinates} \\ m_C = m}} d_C, \qquad 0 \leq m \leq k.$$

A component contributes its degree, rather than just one: it can contain several geometric generic points. In particular,

$$D_0(p) = \sum_{C \text{ exceptional}} d_C.$$

The distinctness of the ordinary coordinates at p ensures that no additional equality partitions are needed outside the two singular clusters.

For $I \subseteq [L]$, let $N_I(p)$ count the points of Γ_p with distinct coordinates that contain the site pairs at every $j \in I$. Additional site pairs are allowed. Then

$$\sum_{\substack{I \subseteq [L] \\ |I|=s}} N_I(p) = \sum_{m=s}^{k} \binom{m}{s} D_m(p). \tag{25}$$

Indeed, a distinct-coordinate solution carrying m site pairs admits exactly $\binom{m}{s}$ choices of s of them.

The auxiliary families used for comparison. For disjoint root-label sets A, B with $|A| = |B| = s$, choose a site subset $I \subseteq [L]$ of size s and bijections

$$f : A \longrightarrow I, \qquad g : B \longrightarrow I.$$

Pin $x_a = 1 - \theta_{f(a)}$ and $x_b = -1 - \theta_{g(b)}$. Divide each remaining original equation by

$$D_t(x_t) = \prod_{j \in I} \big((x_t + \theta_j)^2 - 1\big),$$

which gives a polynomial G_t. For each partition π of the free labels, retain one G_t per block and impose its block equalities. Together with the pinned equations these define the auxiliary family. Its special fiber is exactly $I_{A,B;\pi}$, because the labelings are injective and $D_t|_{\theta=0} = (x_t^2 - 1)^s$. We include all these local partition families in the standing finite-flatness assumption. No matching between A and B is fixed: both bijections are summed over.

Proposition 5.2 (Component formula for the local weights)**.** *Under the finite-flatness and generic reducedness assumptions above,*

$$\boxed{w_s(p) = \frac{1}{n!\binom{L}{s}} \sum_{m=s}^{k} \binom{m}{s} D_m(p), \qquad 0 \le s \le k.} \tag{26}$$

In particular,

$$w_k(p) = \frac{D_k(p)}{n!\binom{L}{k}}.$$

Consequently the local index retains precisely the exceptional components:

$$\boxed{w_p(\tilde{Z}) = \sum_{s=0}^{k} (-1)^s \binom{L}{s} w_s(p) = \frac{D_0(p)}{n!} = \frac{1}{n!} \sum_{C \text{ exceptional}} d_C \ge 0.} \tag{27}$$

Equivalently, the weights satisfy the generating identity

$$n! \sum_{s=0}^{k} \binom{L}{s} w_s(p) t^s = \sum_{m=0}^{k} D_m(p)(1+t)^m.$$

Proof. First, generic simplicity excludes free roots at pinned endpoints. Let q be a generic solution of an auxiliary reduced system. It is also a solution of the original equations, since the equations at the pinned roots vanish automatically. Suppose a free coordinate x_t equals a pinned coordinate x_b at q. The equation P_b^θ vanishes identically after pinning its site pair, so

$$\frac{\partial P_b^\theta}{\partial x_t}(q) = 0.$$

On the pinned locus, $P_t^\theta = D_t G_t$. Since $D_t(q) = G_t(q) = 0$,

$$\frac{\partial P_t^\theta}{\partial x_t}(q) = 0.$$

Permutation symmetry at $x_b = x_t$ gives the same vanishing for the other two entries of the corresponding two-by-two Jacobian block. For every other row, the derivatives in columns b, t agree. Hence

$$DP^\theta(q)(e_b - e_t) = 0,$$

contradicting simplicity of the original generic zero. Thus the free coordinates avoid every pinned endpoint.

Second, depermutation counts the distinct free roots. At such a solution, all D_t are units. The local auxiliary ideal therefore equals the original Bethe ideal together with the pinning equations. Since the original local algebra at q is a field, the auxiliary solution is simple. Imposing any compatible block equalities still gives local length one. If σ is the equality partition of the free coordinates, their unnormalized depermutation weight is

$$\sum_{\pi \le \sigma} \mu(\hat{0}, \pi) = \begin{cases} 1, & \sigma = \hat{0}, \\ 0, & \sigma \neq \hat{0}. \end{cases}$$

Together with endpoint avoidance, this retains exactly the solutions whose full n coordinates are distinct.

Finally, compare degrees and keep track of the labelings. Fix I with $|I| = s$. Every solution counted by $N_I(p)$ has a unique placement of those site endpoints among the ordered root labels. Thus summing the unnormalized auxiliary depermutation counts over all A, B and bijections f, g gives $N_I(p)$. For a fixed A, B, there are $(s!)^2$ pairs of bijections onto I. Every corresponding partition family has central local length $\ell_p(I_{A,B;\pi})$, preserved by flatness. Consequently

$$\begin{aligned} \sum_{|I|=s} N_I(p) &= \binom{L}{s} (s!)^2 \sum_{\substack{A \cap B = \varnothing \\ |A| = |B| = s}} \sum_{\pi \in \Pi([n] \setminus (A \cup B))} \mu(\hat{0}, \pi) \ell_p(I_{A,B;\pi}) \\ &= n! \binom{L}{s} w_s(p), \end{aligned}$$

where the last equality is exactly the normalization in (12). Combining this with (25) proves (26). Taking the alternating sum and using $\sum_{s=0}^{m} (-1)^s \binom{m}{s} = 0$ for $m > 0$ gives (27); the generating identity follows from the binomial theorem. □

The proposition is conditional on finite flatness of the specified injectively labeled auxiliary families and generic reducedness of the full Bethe scheme. It requires no separate generic simplicity assumption for the auxiliary systems. It does not yet establish the physical interpretation of the remaining degree: for $k \ge 2$, the desired vanishing still requires proving that no exceptional component specializes to p.

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The main ideas of this work originate with the author. AI tools were used to assist with writing, language editing, and locating relevant references. The author is responsible for the mathematical content, the accuracy of the references, and the final text.

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